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21 OCTOBER 1959
SUPERSEDING
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10 MARCH 1954

MILITARY SPECIFICATION

LIFE PRESERVERS, VEST, WORK TYPE, UNICELLULAR PLASTIC

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers life preservers with unicellular plastic buoyant pads.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standard, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

V-F-106 — Fasteners, Slide, Interlocking.
V-T-276 — Thread, Cotton.
CCC-C-426 — Cloth, Cotton Drill.
DDD-S-751 — Stitches, Seams; and Stitching.

MILITARY

MIL-P-116 — Preservation, Methods of.
MIL-C-299 — Cloth, Cotton, Print.
MIL-F-495 — Finish, Chemical, Black, for Copper Alloys.
MIL-W-530 — Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-P-3454 — Packaging and Packing of Life Preservers.
MIL-C-10296 — Cloth, Cotton, Sateen, Carded.
MIL-P-15280 — Plastic Foam, Unicellular, Buoyant, Sheet and Molded Shape.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

FSC 4220

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(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The material shall be in strict compliance with all component parts of the finished preserver as specified herein.

3.1.1 *Drill.* The drill shall conform to type 1, class 3 of Specification CCC-C-426 and shall be vat dyed. The dyed drill shall show good fastness to light, salt water, laundering, and crocking.

3.1.2 *Sateen.* The sateen shall conform to Specification MIL-C-10296 except that it may be dyed with napthol colors and shall show good fastness to light, salt water and laundering and air fastness to crocking.

3.1.3 *Color.* The color shall be formula No. 113/49 orange (see 6.2) or olive drab as specified (see 6.1).

3.1.4 *Buoyant pads.* The buoyant pads shall be unicellular plastic material conforming to type IA of Specification MIL-P-15280.

3.1.5 *Webbing.* The webbing shall be cotton webbing equal in color and color fastness to the cotton drill. The 1¼ inch webbing shall conform to type IIa of Specification MIL-W-530 and the ¾ inch webbing shall conform to type I of Specification MIL-W-530.

3.1.5.1 The cotton webbing shall be made mildew resistant and water repellant in accordance with Specification MIL-W-530.

3.1.6 *Metal hardware.* The clasp and slide adjuster shall be brass and shall have a lusterless black finish conforming to type III of Specification MIL-F-495. The clasp shall have a minimum breaking strength of 360 pounds when tested as specified in 4.4.2.

3.2 Manufacture and assembly.

3.2.1 *General.* The manufacture and assembly details of the life preservers shall conform to figures 1, 2, and 3.

3.2.2 *Cover.* The cover shall be made of cotton drill or sateen as specified (see 6.1) and shall be assembled as shown on figure 1.

3.2.2.1 The dull side (filling effect) of the sateen shall face outware on all parts of the finished preserver.

3.2.3 *Shoulder and body straps.* The shoulder and body straps and retaining loops shall be located as shown on figure 1.

3.2.4 *Buoyant pads.* There shall be two front pads and one back pad of unicellular plastic material as specified in 3.1.4 cut from sheet stock or molded. Pads shall be in accordance with figure 3.

3.2.5 *Buoyancy.* The three pads shall have a total buoyancy of not less than 17.5 pounds when tested as specified in 4.4.1.

3.2.6 *Metal hardware.* Metal hardware shall be in accordance with figure 2.

3.2.7 *Slide fasteners.* The slide fastener shall be nylon or nylon tape, have enameled zinc alloy die cast sliders and shall conform to type I, style 3, size LM, class A of Specification V-F-106 except that the tape shall be 9/16 inch. Length of slide fasteners shall be 8 inches for the front panels of the preserver and 10 inches for the pack panel. Fasteners shall pass the salt spray test specified in Specification V-F-106. Fasteners shall have a perforated pull tab. Color of tape shall acceptably match the color of the cotton drill specified in 3.1.1.

3.2.8 *Stitching.* All stitching shall be type 301 of Specification DDD-S-751. Stitching of cotton drill cover shall be done with type IB3, No. 20, 4-ply cotton thread equal in color to the cotton drill and conforming in physical characteristics and color fastness to

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Specification V–T–276. Stitching in webbing and webbing attachments to drill shall be done with type IB3, No. 16 4-ply cotton thread equal in color to the cotton drill and conforming in physical characteristics and color fastness to Specification V–T–276.

3.2.8.1 *Belt strap loops.* Belt strap loops shall be double stitched to the life preserver back as shown on figure 1.

3.2.9 *Tag.* A tag of bleached print cloth in accordance with type II of Specification MIL–C–299, 2½ inches by 3½ inches, shall be sewed on the bottom of the life preserver on the inside of the back cover as shown on figure 1. The printing on the tag shall be in waterproof black ink and shall be as shown on figure 2.

3.2.10 *Slider protection patch.* A piece of drill, 2¾ inches by 4 inches with one long edge selvage, shall be turned under ¼ inch on three sides excluding selvage and hemmed. The patch shall be placed over the open end of the slide fastener and stitched on both inside and outside jacket covers, leaving the edge toward the slider open as indicated in figure 1. A sufficient fullness to provide for easy covering of slider shall be allowed.

3.3 **Workmanship.** The workmanship and finish (such as turnunder of raw ends of webbing) shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 **Lot.** All life preservers offered at one time shall be considered a lot for the purposes of acceptance inspection.

4.3 Sampling.

4.3.1 *Sampling for acceptance inspection.*

4.3.1.1 *Sampling for visual and dimensional examination.* A random sample of finished life preservers shall be selected from each lot of material offered for examination of the materials, construction and workmanship, with lot acceptance based on the sampling examination requirements of table I in accordance with the procedures of Standard MIL–STD–105.

TABLE I. *Sampling for visual and dimensional examination AQL (approx.) = 1.5 percent defective.*

Number of life preservers in lot	Number of life preservers in sample	Acceptance number	Rejection number
2 to 8	All	0	1
9 to 15	8	0	1
16 to 25	10	0	1
26 to 40	13	0	1
41 to 65	17	0	1
66 to 110	22	1	2
111 to 180	28	1	2
181 to 300	35	1	2
301 to 500	45	2	3
501 and over	55	2	3

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4.3.1.2 *Sampling for buoyancy test.* A random sample of finished life preservers shall be selected from each lot by the inspector in accordance with table II and shall be subjected to the test specified in 4.4.1.

TABEE *II. Sampling for tests.*

Number of life preservers of clasps in lot	Number of life preservers in sample	Number of clasps in sample	Acceptance number	Rejection number
110 or under	5	3	0	1
111 to 500	7	5	0	1
501 to 1300	10	7	0	1
1301 to 3200	15	10	0	1
3201 and over	20	15	0	1

4.3.1.3 *Sampling for clasp breaking strenght test.* Prior to attachment of clasps to webbing straps, a random sample of clasps shall be selected, from the lot to be assembled, by the inspector in accordance with table II and shall be subjected to the test specified in 4.4.2.

4.4 **Test procedures.**

4.4.1 *Buoyancy test.* The buoyancy test shall be conducted as follows: Remove pads from the outer cover. Place the two front pads and the back pad in a cage of adequate size and of sufficient weight to submerge all of the pads at one time. This assembly shall be suspended from a direct reading scale in fresh water, at room temperature, with the top of the cage 2-inches beneath the surface of the water. The scale reading shall be taken as soon as practicable and the buoyancy calculated as the apparent weight of the cage and pads subtracted from the previously determined underwater weight of the assembly without the pads.

4.4.2 *Claps breaking strength test.* A section of webbing shall be attached to each side of the engaged clasp. The specimen shall be of sufficient length so that the distance between the jaws of the testing machine shall be approximately 10 inches after the specimen has been clamped for testing. The rate of separation of the jaws of the testing machine shall be approximately 12 inches per minute. The load shall be increased to 360 pounds and held at that value for 1 minute.

4.5 Packaging. Sample packages of life preservers shall be selected from each shipment lot in accordance with Specification MIL–P–116 and shall be inspected to verify conformance to the requirements of 5.1.

5. PREPARATION FOR DELIVERY

5.1 Life preservers shall be packaged by level A or C, packed by level A, B or C, as specified (see 6.1), and marked in accordance with Specification MIL–P–3454.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Color of preserver required (see 3.1.2).

(c) Whether drill or sateen cover is required (see 3.2.2).

(d) Selection of applicable level of packaging and packing (see 5.1).

6.2 Color chip for Formula Number 113/49 can be obtained from Material Laboratory, New York Naval Shipyard, Brooklyn,

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N. Y. When requesting, specify for what purpose color chip is desired.

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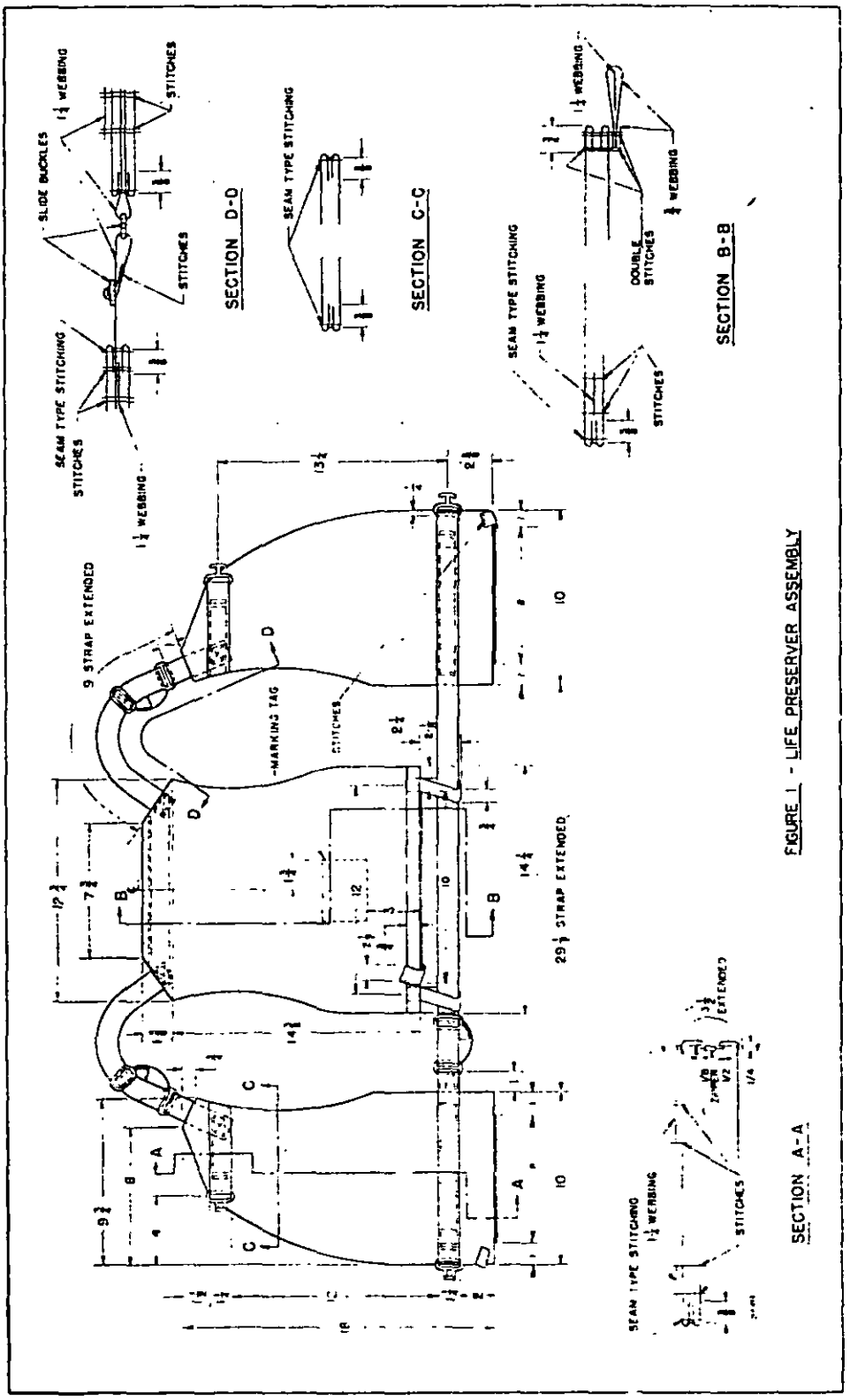


FIGURE 1 - LIFE PRESERVER ASSEMBLY

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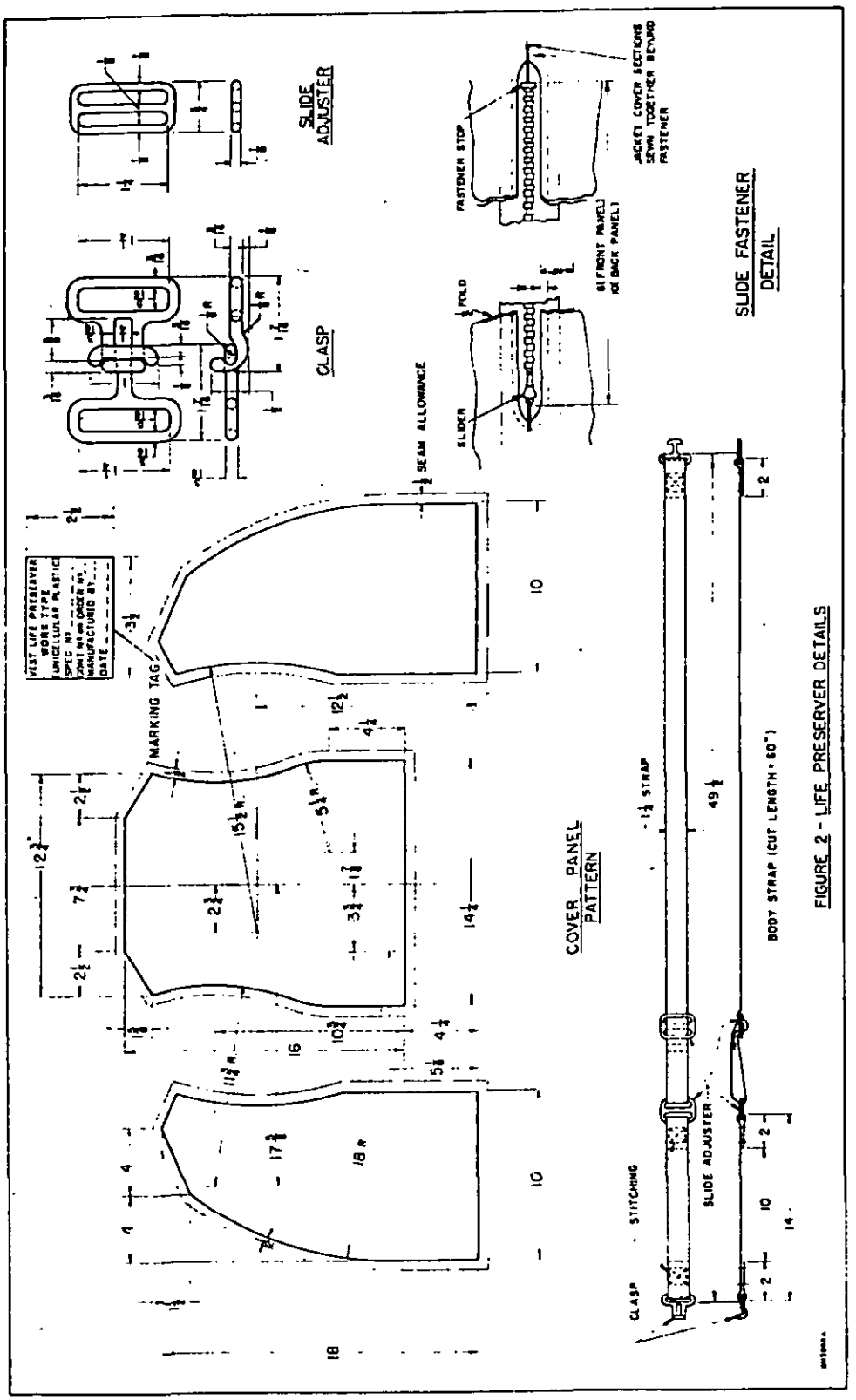


FIGURE 2 - LIFE PRESERVER DETAILS

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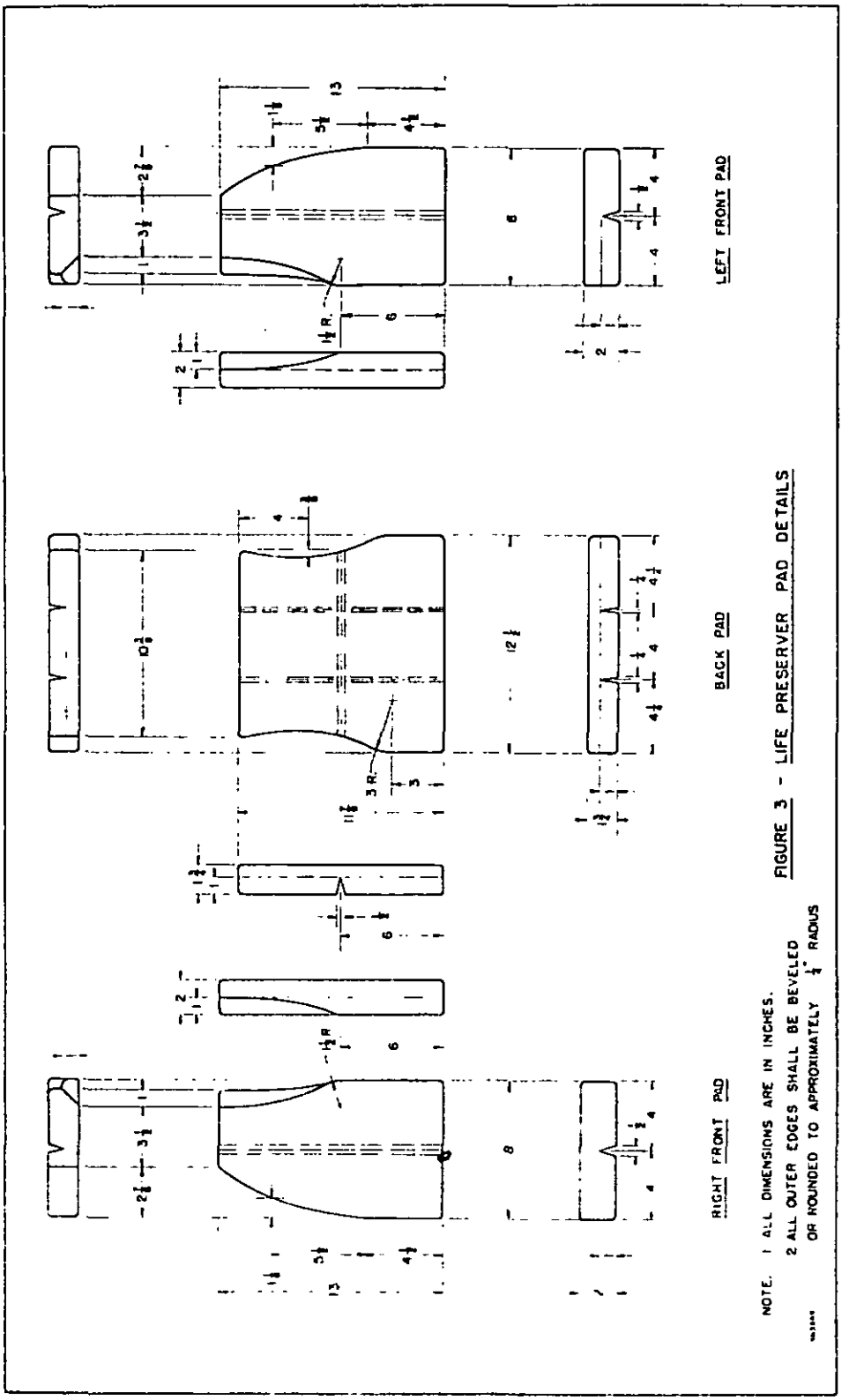


FIGURE 3 - LIFE PRESERVER PAD DETAILS

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